

*MA.8.AR.3.5***Benchmark**

MA.8.AR.3.5 **Given a real-world context, determine and interpret the slope and y-intercept of a two-variable linear equation from a written description, a table, a graph or an equation in slope-intercept form.**

Example: Raul bought a palm tree to plant at his house. He records the growth over many months and creates the equation $h = 0.21m + 4.9$, where h is the height of the palm tree in feet and m is the number of months. Interpret the slope and y-intercept from his equation.

Benchmark Clarifications:

Clarification 1: Problems include conversions with temperature and equations of lines of fit in scatter plots.

Connecting Benchmarks/Horizontal Alignment

- MA.7.AR.4.5
- MA.8.AR.3.2, MA.8.AR.3.3
- MA.8.AR.4.1, MA.8.AR.4.2, MA.8.AR.4.3
- MA.8.F.1.3
- MA.8.DP.1.3

Terms from the K-12 Glossary

- Intercept
- Linear Equation
- Slope

Vertical Alignment**Previous Benchmarks**

- MA.6.AR.3.2

Next Benchmarks

- MA.912.AR.2.5
- MA.912.F.1.5

Purpose and Instructional Strategies

In grade 7, students solved real-world problems involving proportional relationships. In grade 8, students interpret the slope and y-intercept of a two-variable linear equation within a real-world context when given a written description, a table, a graph or an equation. In Algebra 1, students will solve mathematical and real-world problems that are modeled by linear functions and will interpret key features of the graph in terms of the context.

- The purpose of this benchmark is to focus on interpreting the slope and y-intercept in a real-world context using information from a table, graph or written description.
- Students identify the rate of change (slope) and initial value (y-intercept) from tables, graphs, equations or verbal descriptions. Students recognize that if the value $x = 0$ is in a table, the y-intercept is the corresponding y-value. Otherwise, the y-intercept can be found by substituting a point and the slope into the slope-intercept form of the equation and solving for the y-intercept. The slope can be determined by finding the ratio between the change in two y-values and the change between the two-corresponding x-values.

- Using graphs, students identify the y -intercept as the point where the line crosses the y -axis and the slope as the vertical change divided by the horizontal change. In a linear equation, the coefficient of x is the slope and the constant is the y -intercept. Students should have practice with equations in formats other than $y = mx + b$, such as $y = ax + b$ or $y = b + mx$.
- Instruction includes using a variety of vocabulary to make connections to real-world concepts and future courses. To describe the slope, one can say either “the vertical change divided by the horizontal change” or “rise over run.”
- In contextual situations, the y -intercept is generally the starting value or the value in the situation when the independent variable is 0.
- The slope is the rate of change that occurs in the problem. Rates of change can often occur over years. In these situations, it is helpful for the years to be "converted" to the number of years since the start year.
 - For example, the years of 1960, 1970, and 1980 could be represented as 0 for 1960, 10 for 1970 and 20 for 1980.
- Students use the slope and y -intercept to write a linear function in the form $y = mx + b$.
- Students should remember to interpret the line of fit within the context of the data provided by the scatter plot (MA.8.DP.1.3). The line of fit is meant to understand the general trend of data, but it might not be able to explain everything about it.
- For mastery of this benchmark, it is not the expectation to compare slopes or y -intercepts of two linear equations in two variables.
- Instruction includes learning about linear relationships within other content areas. Students should recognize that the conversion between Fahrenheit and Celsius represents a linear relationship, but not a proportional one. Memorization of the formulas is not an expectation of the benchmark.
 - The formula for converting Fahrenheit to Celsius is: $C = \frac{5}{9}(F - 32)$.
 - The formula for converting Celsius to Fahrenheit is: $F = \frac{9}{5}C + 32$.

Common Misconceptions or Errors

- Students may incorrectly identify the slope and y -intercept.
- Students may incorrectly interpret the slope and y -intercept.
- The misconceptions of this benchmark may develop for some students based on the real-world context of the problems presented. To address this misconception, scaffold questions to help students understand the context.

Strategies to Support Tiered Instruction

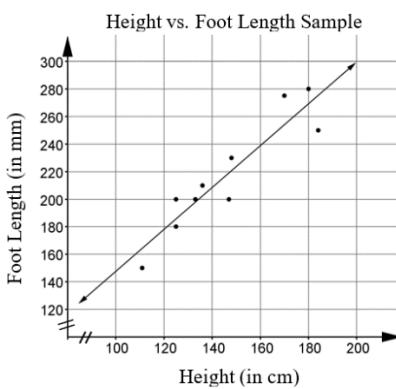
- Teacher provides opportunities for students to comprehend the context or situation by engaging in questions.
 - What do you know from the problem?
 - What is the problem asking you to find?
 - What are the two quantities in this problem?
 - How are the quantities related to each other?
 - Which quantity do you want to consider as the independent variable?
 - Which quantity do you want to consider as the dependent variable?

- Instruction includes the use a three-read strategy. Students read the problem three different times, each with a different purpose.
 - First, read the problem with the purpose of answering the question: What is the problem, context, or story about?
 - Second, read the problem with the purpose of answering the question: What are we trying to find out?
 - Third, read the problem with the purpose of answering the question: What information is important in the problem?
- Instruction includes the understanding that the independent variable depends on the given context. Additionally, independent variables are not always the x -axis and the dependent variable are not always the y -axis.
 - For example, if one has a proportional relationship between feet and meters, they can graph feet either on the x -axis or the y -axis. Which one that is dependent depends on the context. For instance, if one is given feet and converting to meters, then feet would be independent and meters would be dependent.

Instructional Tasks

Instructional Task 1 (MTR.7.1)

The graph below shows a scatter plot and its line of fit for data collected on the height and foot length of a sample of 10 male students.



Part A. What does the graph indicate about the relationship between foot length and height?

Part B. The equation of the line of fit is $f = 1.5h - 4.3$, where f is foot length in millimeters and h is height in centimeters. Explain the meaning of the slope and the f -intercept of this equation in the context of the data.

Instructional Items

Instructional Item 1

At Stay-a-While Coffee shop, they display their internet fees on a chart like the one shown below. Determine the slope for the relationship between the number of minutes, x , and the amount charged, y .

Time	Amount Charged
1 hour	\$3.95
2 hours	\$6.95
4 hours	\$12.95

Instructional Item 2

Joshua adopted a puppy from a dog shelter. He records the puppy's height over many months and creates the equation $h = \frac{m}{5} + 3$, where h is the height of the puppy, in feet, and m is the number of months. Interpret the slope and h -intercept from his equation.

**The strategies, tasks and items included in the BIG-M are examples and should not be considered comprehensive.*